

DINGYUN HUANG

E-mail: dr.dingyun.huang@gmail.com
Mobile: +44 7398990697

EDUCATION

2021 – 2026	University of Cambridge – PhD in Physics Data-driven methods for materials discovery applications in optoelectronics
2017 – 2021	University of Cambridge – BA & MSci in Natural Sciences 1 st class honours with specializations in Physics
2015 – 2017	Lime House School – A Levels

SCHOLARSHIPS & AWARDS

2025	Royal Society of Chemistry Materials Chemistry Horizon Prize
2021 – 2025	Cambridge Trust PhD Scholarship at the University of Cambridge
2017 – 2021	Churchill College Scholarship for outstanding performance in Tripos examinations
2016 – 2017	Top Gold Awards (UK top 50) in the British Physics and Astronomy & Astrophysics Olympiads

SELECTED PUBLICATIONS

Huang, D.; Cole, J. M. Machine-Learning Predictions of Photoluminescence in Molecules Exhibiting Thermally Activated Delayed Fluorescence with Implicit Experimental Validation. *J. Chem. Inf. Model.* 2026, 66 (10), 5757–5763.

Huang, D.; Cole, J. M. Cost-Efficient Domain-Adaptive Pretraining of Language Models for Optoelectronics Applications. *J. Chem. Inf. Model.* 2025, 65 (5), 2476–2486.

Huang, D., Cole, J.M. A database of thermally activated delayed fluorescent molecules auto-generated from scientific literature with ChemDataExtractor. *Sci Data* 11, 80 (2024).

Zhu, Z.; Song, P.-B.; Huang, D.-Y.; Xu, H.-K.; Gu, Y.-S.; Yu, Q.-Y.; Liu, D.; Xue, S.-S.; Guan, D.-D.; Wang, S.-Y.; Li, Y.-Y.; Liu, C.; Shi, Y.-G.; Xu, X. Y.; Zheng, H.; Jia, J.-F. Evolution of One-Dimensional Electronic Dynamics in $\text{Nb}_{2n+1}\text{Si}_n\text{Te}_{4n+2}$ *Phys. Rev. B* 2022, 106 (8), 085426.

RESEARCH PROJECTS

University of Cambridge – Research Associate

2025-2026	Optimization of galvanizing line control with deep learning with Tata Steel UK - Delivered 1% fractional-error coating weight predictions by building an end-to-end deep learning pipeline for noisy and biased production signals into decision-ready inputs for real-time line control. - Corrected coating-weight deviations within ten time-steps by deploying a real-time feedback model for production-parameter optimization.
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University of Cambridge – PhD in Physics

2024 – 2025	End-to-end workflow of training machine learning predictors for PL wavelengths - Curated a high-precision dataset of experimental PL wavelengths from literature, exceeding 90% extraction precision consistently. - Expanded usable training data by 20% by unlocking reliable extraction from Wiley articles PDFs. - Achieved 0.12 eV MAE for TADF PL wavelength prediction with a heterogeneous GNN model.
2023 – 2024	Cost-efficient fine-tuning of large language models for optoelectronics - Delivered three language models for optoelectronics tasks with continued pre-training and fine-tuning, including data-mining through multi-turn Q&A, and text embedding. - Demonstrated RoBERTa-adapted models required 80% less continued pre-training than BERT to reach comparable performance in optoelectronics, establishing a cost-efficient domain-adaptation strategy for specialized scientific language models.

2021 – 2023	Automatic database collection for TADF materials through text-mining - Achieved 82% precision in curating an OLED materials database by devising an automated text-mining pipeline basing on ChemDataExtractor. - Drove 800+ database downloads and enabled reuse across three peer-reviewed journal articles. - Raised the chemical name assignment precision from 30% to 80% by innovating a statistical compound-name screening method for high-confidence database construction and curation. - Delivered a 92%-accurate OPSIN-based structure-resolution workflow from journal text, helping PyOPSIN reach ten thousand annual PyPI downloads across the computational chemistry community.
2021 - present	ChemDataExtractor - Increased data extraction recall by 30% by resolving critical ChemDataExtractor failures across abbreviation detection, text/table parsing, and XML scraping modules. - Maintained ChemDataExtractor2 production readiness by resolving user-reported GitHub issues. - Modernized ChemDataExtractor from AllenNLP to Torch/HuggingFace while updating software support from Python 3.6 to Python 3.11 for broader scientific software compatibility. - Reduced installation time from hours to minutes by systemizing required dependencies substantially.

WORK EXPERIENCES

2025 – 2026	Department of Physics, University of Cambridge – Research Associate Delivered 1%-error zinc-coating prediction for steel galvanization and developed feedback optimization models to reduce coating deviations and zinc waste.
Summer 2025	Huawei Hong Kong Research Centre, Theory Lab – Research Scientist Internship Reduced regression errors by 65% and trainable parameters by 90% through physics-informed neural-network innovation for optical-imaging optimization. Accelerated data loading and reduced memory overhead by implementing scalable sharding methods.
2024 – 2025	Churchill College, University of Cambridge – Mathematics Supervisor Simplified advanced mathematical concepts and improved teaching of problem-solving skills. Fostered a supportive and effective learning environment for six Natural Sciences students in learning Mathematics, five out of six of my students achieved first class in their examination.
Apr – Sep 2022	University of Cambridge STEM Smart Project – Supervisor Improved university readiness for underrepresented students by designing interactive mathematics lessons that closed knowledge gaps and built confidence in advanced STEM topics.
Jan – Mar 2022	University of Cambridge – Python Computing Demonstrator Improved students' programming proficiency by teaching numerical physics methods through tailored exercises matched to different skill levels.

ACADEMIC PRESENTATIONS

2025	Poster at AI for Science Summit, Accelerate Programme, University of Cambridge
2025	Oral Presentation at Conference on Everything, Churchill College, University of Cambridge
2024	Poster at 7th Machine Learning and AI in Bio(Chemical) Engineering Conference

TECHNICAL SKILLS

- ▶ **Python:** PyTorch, Scikit-Learn, Pandas, and other popular data-science/ML packages; large-scale data manipulation and visualization; object-oriented programming and code packaging for managing ChemDataExtractor and group websites.
- ▶ **High performance computing:** exascale supercomputing clusters; PBS job scheduling; MPI-based multi-process workflows; multi-GPU and multi-node distributed model training; website front and back end through AWS.
- ▶ **Machine Learning:** CNN, GNN, Transformers, LSTM, Random Forests, Bayesian methods, feature engineering.
- ▶ **Language Modelling:** conventional NLP methods; pre-training and fine-tuning transformer-based language models, such as BERT and Llama; deploying language models on downstream applications such as knowledge extraction through multi-turn QA and RAG. Agentic workflow for scientific discoveries.
- ▶ **Material modelling:** cheminformatics toolkit RDKit; Cambridge Structural Database; property modelling using graph neural networks and other ML algorithms.
- ▶ **Git & PyPI:** version management and collaboration on group software through GitHub; package publishing via PyPI.